Shifting from plant functional types to traits: Enhancing biodiversity representation in land surface models

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Open Research Statement

Data are already published and publicly available, with those items properly cited in this submission. The datasets used in the study are accessible at Caltech Data Library under https://doi.org/10.22002/7xgrn-qtc49, including: the SHIFT AVIRIS-NG dataset for the

Dangermond Preserve at 30 m resolution, which offers detailed spectral reflectance information across 399 wavelengths, TROPOMI sun-induced fluorescence at 740 nm, CliMA Land model outputs of ecosystem fluxes, trait variations for different plant functional types, LiDAR-derived clumping index, and PLSR-derived traits. These datasets are available for the period of February to May 2022. The Level 2A (L2A) unrectified surface reflectance images from NASA's Airborne Visible / Infrared Imaging Spectrometer-Next Generation (AVIRIS-NG) instrument collected as part of the Surface Biology and Geology High-Frequency Time Series (SHIFT) campaign including reflected radiance at 5-nm intervals in the Visible to Shortwave Infrared (VSWIR) spectral range from 380-2510 nm can be found at https://doi.org/10.3334/ORNLDAAC/2183 (Brodrick et al., 2023). The California Multi-Source Vegetation Layer, specifically depicting Wildlife Habitat Relationship classes (WHRTYPE) (California Department of Forestry and Fire Protection) can be found at https://gis.data.ca.gov/maps/CALFIRE-Forestry::california-vegetation-whrtype/about. The CliMA Land model used to simulate surface hyperspectral reflectance and transmittance, energy and carbon fluxes is available at https://github.com/CliMA/Land. The code used in the analysis presented in this paper is available on GitHub and can be accessed at https://github.com/braghiere/shift_dangermond_trait/releases/tag/v0.3.0-ecosphere-revision2.

Abstract

Accurate representation of the interplay between ecosystems and climate in Earth system models (ESMs) is important for understanding and predicting changes in carbon, water, and energy cycles. The traditional land components of ESMs, the Land Surface Models (LSMs), often rely on the categorization of plant functional types (PFTs), a method that may overlook the ecological variations within plant groups. Using the CliMA Land model with surface hyperspectral data sourced from AVIRIS-NG (Airborne Visible/Infrared Imaging Spectrometer-Next Generation) during the NASA's SHIFT (Surface Biology and Geology High-Frequency Time Series) campaign, we derive leaf biogeochemical traits and compare a trait-based land modeling approach to the commonly used PFT one. Our findings reveal significant spatial and temporal variations in traits and other related optical and photosynthetic parameters, not only within the same PFT but also across different seasons. Validation against the TROPOspheric Monitoring Instrument (TROPOMI) relative solar-induced fluorescence at 740 nm (SIF_{740}) showed comparable R^2 values of 0.68 (PFT) and 0.69 (Trait), while the trait-based approach better represented phenology and spatial heterogeneity. Our findings highlight the limitations of traditional PFT-based approaches in LSMs and underscore the importance of incorporating detailed, trait-based data for a more accurate and comprehensive understanding of ecosystem functions.

Keywords: Biodiversity; CliMA Land model; Ecosystem Dynamics; Imaging Spectroscopy; Land Surface models; Photosynthesis; Plant Functional Types; Plant Traits; Solar-Induced Fluorescence.

1 Introduction

The global carbon sink is significantly influenced by the terrestrial carbon uptake, which, despite its high interannual variability (Le Quéré et al., 2009; Piao et al., 2020) and associated uncertainty (Randerson et al., 2025), offsets approximately one-third of anthropogenic CO_2 emissions (Friedlingstein et al., 2023). Therefore, the Earth's land surface, with its mosaic of nearly 435,000 plant species (Enquist et al., 2019), is crucial to the global climate system. This vast biodiversity, including functional (trait) diversity (McGill et al., 2006; Violle et al., 2007), introduces substantial uncertainty into land processes and climate projections, highlighting the need to enhance their representations in the land components of Earth system models (ESMs), the Land Surface models (LSMs) (Braghiere et al., 2023a; Braghiere et al., 2023b). Incorporating fundamental biology, such as phenotypic diversity, into LSMs is essential for its intrinsic value and global policy relevance, but particularly in conservation and land management. Functional diversity also mediates ecosystem responses to disturbances (e.g., fire, drought, pests, and land-use change), which remains one of the most challenging aspects of current ESMs (Meinshausen et al., 2024), and could potentially alter the capacity for net-negative carbon emissions needed to mitigate climate change.

The challenge lies in measuring, predicting policy outcomes, and management impacts on ecosystem functions, where models can play a crucial role if they accurately represent functional diversity and community dynamics. While models agree that atmospheric CO_2 and nitrogen deposition mainly drive gains in terrestrial carbon stocks, several unique yet interconnected mechanisms govern the movement of carbon into, within, and out of terrestrial ecosystems, including photosynthesis, respiration, carbon allocation, plant growth, litterfall, and mortality (O’Sullivan et al., 2022). Introduced in the 1980s, the ‘greening’ of global climate models or ESMs (Farquhar et al., 1980; Sellers et al., 1997) attempted to depict the vast diversity of the land surface through the lens of plant functional types (PFTs), which aim to simplify the representation of terrestrial vegetation by classifying it into discrete groups with shared functional characteristics (Bonan et al., 2002). The coarse discretization inherent in the PFT-approach often overlooks the nuanced variations and dynamic interactions within ecosystems and it is an area of ongoing debate (Cranko Page et al., 2024; Zakharova et al., 2019). Moreover, the effectiveness of PFTs in accurately representing vegetation variation and in translating climate inputs into flux outputs for ecosystems is increasingly questioned (Anderegg et al., 2022; Dietze et al., 2014; Harrison et al., 2021; Raczka et al., 2018; Wang & Braghiere et al., 2025). A growing body of research advocates for a more granular representation of vegetation in LSMs (Alton, 2011; Reichstein et al., 2014), suggesting trait-based approaches for a more detailed and accurate portrayal of plant characteristics. Additionally, LSMs often struggle to incorporate more realistic biology, finding their structures inadequate when attempting to integrate traits (Butler et al., 2022). However, despite these suggestions, the use of PFTs persists in analyzing terrestrial fluxes within LSMs partly due to the convenience of PFTs in a data-poor world, allowing for simpler data interpretation and extraction of generalizable patterns from complex datasets (Lovenduski & Bonan, 2017; Cranko Page et al., 2022).

The PFT concept, while useful, does not accurately reflect how plants actually function or the ecological meaning of their associated traits, which are more closely linked to the underlying genetics and their environmental responses. PFT-based models rely on static values from look-up tables rather than modeling the dynamic physiology of different functional types, which introduces significant limitations. Recent data reveal the phenology of traits assumed to be static and highlight within-species heterogeneity rather than PFT-level homogeneity (Chadwick et al., 2025; Kattge et al., 2011), illustrating the shortcomings of this approach. For example, the Joint UK Land Environment Simulator (JULES) (Best et al., 2011; Clark et al., 2011) demonstrated improved performance when four additional PFTs were added to the original five (Harper et al., 2016), enhancing its ability to accurately replicate global vegetation distribution and function. While having a unique set of parameters for each plant species would be ideal yet impractical, determining the optimal number of PFTs remains an open question.

Although PFTs offer a way to simplify and manage complexity and parametric uncertainty in LSMs (Shiklomanov et al., 2019), moving towards more realistic, trait-based models could potentially reduce uncertainties by using actual physical trait values instead of highly generalized estimates that are often outdated and based on localized experiments. Incorporating trait diversity and remote sensing variables into models has seen various attempts with notable limitations. For instance, the Lund-Potsdam-Jena General Ecosystem Simulator (LPJ-GUESS) (Smith et al., 2014) uses a trait-based approach but does not allow direct input or model updates of traits, instead characterizing trait distributions for PFTs. The Ecosystem Demography (ED)2 model (Medvigy et al., 2009) can initialize structure from remote sensing data but struggles to incorporate physiological differences, often updating PFTs by their trait means. In the Forest Model Individual-Based (FORMIND) model (Fischer et al., 2016), remote sensing data such as vegetation structural diversity from the Global Ecosystem Dynamics Investigation (GEDI) mission (Dubayah et al., 2020; Hancock et al., 2019; Tang & Armston, 2019) have been used to update forest structure, yet direct assimilation of these measurements remains challenging. These examples highlight the ongoing efforts and significant challenges in integrating trait variability into models, underscoring the need for improved methodologies and frameworks.

Despite the recent availability of large-scale databases that enable more refined modeling, the practical application of these data within LSMs is challenged by several factors (Fisher & Koven, 2020). First, the integration of high-resolution trait data into existing LSMs often requires significant adjustments in model structure and parameterization (Fisher et al., 2015; Medlyn et al., 2011). This includes recalibrating model responses to account for varied plant behavior under different environmental conditions, a process that can complicate model validation and sensitivity analyses. Additionally, trait data often exhibit substantial variability both within and between species, adding layers of complexity to their representation in models (Wright et al., 2004) and often requiring higher resolution and extra computing power.

Despite the existence of models with environmentally determined levels of traits like chlorophyll, nitrogen, and lignin that have worked effectively (Botkin et al., 1972; Bugmann & Seidl, 2022), PFTs have persisted as a convenient shorthand for modelers lacking the data necessary for more biologically realistic equations. Additionally, the scale at which these traits influence ecosystem processes often exceeds the spatial and temporal resolution of available observations, necessitating the use of upscaling methods that may introduce additional uncertainties (Smith et al., 2001). Thus, the gap between trait databases and their practical application in LSMs highlights the ongoing challenges in model development. Moreover, these trait databases themselves often have significant gaps, particularly in regions where many traits have not yet been measured or derived together, which limits the ability to establish relationships between traits (Cornwell et al., 2019).

The lack of repeated measurements, due to the non-systematic nature of efforts like the TRY database (Kattge et al., 2011), further complicates the modeling process. However, spaceborne imaging spectroscopy (e.g., EMIT (Green et al., 2020; Thompson et al., 2019, 2024), EnMAP (Guanter et al., 2015), PRISMA (Cogliati et al., 2021), and PACE (Werdell et al., 2019)) offers a promising solution to these issues by providing systematic and repeated measurements of plant traits across large spatial scales. This technology can help overcome the limitations of current trait databases and enhance the accuracy and applicability of trait-based approaches in LSMs.

Arising from the established need to analyze and eventually surpass the limitations of PFTs in LSMs, this study presents a novel approach using the CliMA Land model (Braghiere et al., 2021; Wang et al., 2021, 2023) to evaluate traditional PFT-based approaches in comparison to a trait-based approach, employing data derived from the AVIRIS-NG (Airborne Visible/Infrared Imaging Spectrometer-Next Generation) Level-2 surface reflectance (native ~5 m) during NASA's SHIFT (Surface Biology and Geology High-Frequency Time Series) campaign conducted from February to May 2022 over Dangermond Preserve in California, USA (Figure 1). High-resolution (30 m) trait maps of leaf chlorophyll content per area (CHL), leaf mass per area (LMA), and leaf water content (LWC) were integrated into the CliMA Land model to investigate their influence on critical leaf optics and photosynthesis parameters (Farquhar et al., 1980; Sellers, 1985), and predictions of energy and carbon fluxes. This approach not only allows us to simulate gross primary production (GPP) and solar-induced fluorescence (SIF) with enhanced precision but also enables a direct comparison between the spatially explicit, trait-informed maps and the conventional PFT representations with parameters fixed in a look-up table format.

In this study, we address whether high-resolution plant trait data integrated into the land component of an ESM can better predict SIF at 740 nm (SIF_{740}), a proxy of productivity, and how it impacts carbon uptake calculations compared to the traditional PFT-approach. Our hypothesis posits that trait-based approaches, informed by detailed, spatially explicit data from the SHIFT campaign, provide an improved framework for modeling complex ecological interactions and carbon dynamics compared to conventional PFT methods. We evaluate this hypothesis by examining the specific research questions:

1. Can the CliMA Land model, combined with hyperspectral data, accurately derive plant traits through inverse modeling, and how do these traits vary across space, time, and PFTs?
2. How do the temporal and spatial variations in CHL, LMA, and LWC compare between the PFT and trait-based approaches?
3. How do predictions of SIF_{740} and GPP with CliMA Land forward modeling validate against the trait-based and the PFT-based approaches, and how well do these approaches validate against TROPOMI data?

The necessity of employing campaigns like SHIFT and LSMs such as CliMA Land for this type of research arises from their potential to provide the detailed, high-frequency data required to operationalize and validate trait-based modeling approaches. This highlights the need to offer systematic, repeated measurements of plant traits across large spatial scales, addressing significant gaps in current trait databases and land modeling approaches. We aim to demonstrate how integrating finely resolved, dynamic plant trait data into LSMs can refine our understanding of terrestrial biosphere processes and enhance the predictive accuracy of these models.

2. Materials and Methods

2.1 Study Area: Dangermond Preserve, California, USA

The Jack and Laura Dangermond Preserve, spanning 9,903 hectares at the juncture of the Northern and Southern California Current marine ecoregions (34.49°N, 120.44°W), is a key conservation area that also bridges the Southwestern and Central Coastal floristic regions of California, USA (Figure 1). This preserve is an area of rich biodiversity, with coast live oak (*Quercus agrifolia*) woodlands covering about a third of its area and serving as a prime example of its diverse ecosystems (Sousa et al., 2022). The preserve's varied elevation, from sea level to 645 meters, along with its complex topography, results in distinct microclimates and varied solar radiation patterns across its expanse. Influenced by a marine inversion layer that brings frequent fog and moderate temperatures, especially to coastal zones, Dangermond Preserve's climate is further characterized by an elevation-dependent precipitation gradient, enhancing its mesoclimatic complexity. The Dangermond Preserve has a Mediterranean-type climate with cool, wet winters and warm, dry summers, with most precipitation falling in late fall–winter. Vegetation activity typically peaks in spring following winter rains and declines into summer as soils dry. The geological and soil diversity, especially the predominance of coast live oak woodlands on specific geologic formations, directly influences the distribution of vegetative communities. Additionally, the preserve's fire history and severe drought periods have impacted its ecological dynamics, vegetation health, and composition, highlighting the intricate interactions between its biotic and abiotic elements. Dangermond Preserve has experienced fewer large wildfires (3) over the last century (< 10% of the total area, see Appendix S1: Figure S1) (The Nature Conservancy, 2025). At the same time, the Preserve is actively managed with prescribed fire to reduce hazardous fuels and restore native communities; recent prescribed burns have been conducted on targeted units within the Preserve (< 1% of the total area).

2.2 Spectral data and LiDAR collection

The NASA’s SHIFT campaign (Chadwick et al., 2025), focusing on the Dangermond Preserve in Santa Barbara County, California, utilized AVIRIS-NG to capture weekly high-resolution (~5 m) spectral imagery over a 13-week period from late February to May 2022. This 13-week window spans late wet-season through spring green-up and the onset of the dry season (late February–May), thereby capturing the rising limb and peak of seasonal activity. This initiative aimed to explore the dynamic interplay between terrestrial vegetation and coastal ecosystems by examining changes in vegetation phenology and ecosystem function through high-frequency, high-fidelity spectroscopic data. By providing unprecedented detail on the subseasonal variations within the Dangermond Preserve’s unique coastal, marine, and terrestrial habitats, the AVIRIS-NG measurements have been instrumental in enhancing our understanding of ecosystem processes, aiding in conservation efforts, and supporting NASA's Surface Biology and Geology (SBG) mission's objectives in scientific advancement and applications development. The campaign's comprehensive data collection underscores the potential of repeat imaging spectroscopy to offer insights into ecological changes, emphasizing the critical role of high-resolution spectral data in ecological research and management.

The dataset generated by AVIRIS-NG during the SHIFT campaign provides a detailed time series of the Earth's surface through Level 2A unrectified surface reflectance images (Brodrick et al., 2023), or the hemispherical-directional reflectance factor (HDRF) (Schaepman-Strub et al., 2006). These images capture the landscape in the Visible to Shortwave Infrared (VSWIR) spectral range from 380 nm to 2510 nm at 5-nm intervals, offering high-resolution imaging spectroscopy data. The AVIRIS-NG sensor achieves this with a 1 milliradian instantaneous field of view, translating to ground sampling distances that can vary from 20 m to sub-meter levels depending on flight altitude. During the SHIFT campaign at Dangermond the flight altitudes produced ~5 m native pixels, which were subsequently resampled to 30 m using area-weighted conservative remapping for modeling (see Appendix S1 for variogram analysis). Upscaling the spatial resolution was used to reduce compute and storage demands. Running spectral fitting on the native 5 m grid over the full Dangermond area would have been prohibitively slow and storage intensive. The dataset supports a range of applications from vegetation monitoring to geological mapping, underpinning studies of ecosystem functions, seasonal variations, and environmental change with unprecedented detail and frequency.

The LiDAR data for the Dangermond Preserve were collected through an airborne survey conducted by Aeroptic LLC in collaboration with The Nature Conservancy and ESRI from September 22nd, 2018, to October 1st, 2018. The survey produced a high-density point cloud dataset comprising over 7.8 billion points with an average point density of 60.68 points per square meter, covering approximately 129.13 km^2. The dataset includes classifications such as ground, buildings, low points, water, road surfaces, and more, facilitating diverse ecological and conservation applications (The Nature Conservancy, 2022). The gap fraction (P_{gap}) was estimated at 5-m resolution directly from LiDAR returns, classified into ground and non-ground, with a 1-m digital terrain model generated via Delaunay triangulation of ground-classified points and used to normalize elevations to height above ground (Ferraz et al., 2012). P_{gap} was then computed from the normalized discrete returns for each grid cell using the Beer–Lambert/Poisson gap-probability framework (Armston et al., 2013) similarly to the one used by the GEDI L2B algorithms (Hancock et al., 2019; Tang & Armston, 2019) without requiring waveform synthesis.

While the LiDAR acquisition predates the SHIFT imaging spectroscopy campaign, the Dangermond Preserve is a protected area where minimal changes in vegetation structure are expected due to the slow growth rates of the dominant vegetation and low disturbance frequency. This stability was further confirmed by examining the Hansen Global Forest Change dataset for the region, which showed no detectable forest loss between 2018 and 2022 (Hansen et al., 2013).

2.3 Plant Functional Types

The California Multi-Source Vegetation Layer, specifically depicting Wildlife Habitat Relationship classes (WHRTYPE) (California Department of Forestry et al., 2015, 2018) plays an important role in offering a comprehensive view of habitat types across California. This dataset, managed by the California Department of Forestry and Fire Protection (CALFIRE) in collaboration with the California Department of Fish and Wildlife VegCamp program and leveraging data from the USDA Forest Service Region 5 Remote Sensing Laboratory, represents the most accurate spatial distribution of habitat types available for the state. In this study, the WHRType field was used as represented in the generalized WHR13 legend (13 statewide classes).

This dataset offers insights into the diverse vegetation cover and PFTs found throughout California. The inclusion of WHRTYPE data categorizes the landscape into distinct habitat classes, each corresponding to specific plant communities and wildlife associations. Such detailed classification aids in identifying areas of ecological significance, prioritizing regions for conservation efforts, and guiding land management practices to preserve biodiversity and ecosystem health.

The Dangermond Preserve embodies a mosaic of PFTs that reflect its unique geographical and climatic conditions. Within the Preserve, there are three predominant PFTs (**Figure 1**) as defined by the WHRTYPE/WHR13 data:

PFT 1. Coastal Scrub, characterized by open, low-lying vegetation, dominated by soft-leaved, often drought-deciduous shrubs (e.g., *Artemisia californica*, *Salvia leucophylla*). At the Preserve, this PFT includes multiple coastal sage scrub alliances. PFT 1 covers ~33% of the area.

PFT 2. Annual Grassland, dominant in the lower elevations of the Dangermond Preserve, is marked by a seasonal cycle of growth and senescence. It consists mainly of non-native grasses that thrive in the Mediterranean climate of California. This PFT is significant for its rapid growth during the wet season, which contributes to carbon sequestration, and its quick decay, which recycles nutrients back into the soil. Despite being a non-native PFT, annual grasslands have become an integral part of the landscape. PFT 2 covers ~38% of the area.

PFT 3. Coastal Oak Woodland, recognized by the prevalence of oaks, particularly the coast live oak, which forms dense canopies that modulate the microclimate beneath them. The leaf litter from oaks enriches the soil, while the trees themselves are crucial for carbon storage. The deep roots of oaks play an important role in water regulation. PFT 3 covers ~29% of the area.

The dynamism within these PFTs, driven by seasonal changes and disturbances such as fires and droughts, challenges the static nature of PFT classifications in modeling and calls for a more nuanced, trait-based approach to truly capture the variability and resilience of these ecosystems.

2.4 The CliMA Land Model

The CliMA Land model, developed by the Climate Modeling Alliance (CliMA), is a next-generation LSM that integrates hyperspectral canopy radiative transfer, soil-plant hydraulics, stomatal optimization, and photosynthesis (Braghiere et al., 2021; Wang et al., 2021, 2023). Unlike traditional LSMs operating at broadband wavelengths, this hyperspectral version of CliMA Land simulates hyperspectral radiative transfer from 400 to 2500 nm at 10-nm resolution, enabling direct comparison with imaging spectroscopy observations.

The model's radiative transfer scheme derives from mSCOPE (Yang et al., 2017), incorporating Fluspect (Vilfan et al., 2016) for simulating leaf-level reflectance, transmittance, and fluorescence, and a formulation based on SAIL (Verhoef, 1984) for canopy-scale radiative transfer. Key modifications in CliMA Land include: (a) explicit inclusion of carotenoid light absorption for accurate calculation of absorbed photosynthetically active radiation (400-700 nm) (Wang et al., 2021, 2023; Wang & Frankenberg, 2022), and (b) incorporation of a clumping index to account for horizontal canopy structure (Braghiere et al., 2021).

Photosynthesis follows the Farquhar-von Caemmerer-Berry model (Farquhar et al., 1980), with maximum carboxylation rate (Vcmax25) and electron transport rate (Jmax25) dynamically linked to leaf chlorophyll content following (Croft et al., 2017):

$$V_{cmax\,25} = 1.30 \cdot CHL + 3.72 \qquad (1)$$

$$J_{max\,25} = 2.49 \cdot CHL + 10.80 \qquad (2)$$

Where CHL is given in $\mu g.cm^{-2}$ and derived from hyperspectral data inverse modeling (Section 2.5).

For this study, CliMA Land was run in offline mode at 30-m spatial resolution as independent snapshot simulations for each of the 13 AVIRIS-NG acquisition dates, using date-specific retrieved traits and ERA5 meteorological forcing (Hersbach et al., 2020) at local noon (19:00 UTC). Soil parameters were derived from Dai et al. (2019) for hydraulic properties and from a soil color classification (Braghiere et al., 2023b; Lawrence & Chase, 2007) for hyperspectral soil background reflectance. These datasets are accessible through GriddingMachine (Wang et al., 2022).

A comprehensive technical description of the model, including detailed specifications of the radiative transfer scheme, canopy structure representation, and numerical implementation, is provided in Appendix S1: Section S1. A big-leaf two-band version of the model designed for calibration and GPU acceleration (Deck et al., 2026) is maintained in a separate repository (https://github.com/CliMA/ClimaLand.jl).

2.5 Deriving Plant Traits from Hyperspectral Surface Reflectance

The inverse modeling process implemented in this study aims to derive vegetation trait parameters from remote sensing hyperspectral surface reflectance data, utilizing the CliMA Land model. For each acquisition date, the inverse modeling workflow was applied independently to retrieve trait parameters from observed surface reflectance at that specific time point. This methodology begins with the configuration and initialization of the model, where specific parameters necessary for the simulation are set. This configuration includes parameters such as soil properties (e.g., soil moisture, soil color, and soil texture), plant physiological traits (e.g., stomatal conductance, leaf nitrogen content, leaf optical properties, and root depth), and atmospheric conditions (e.g., temperature, humidity, and incoming radiation). Additionally, the SIF component is disabled, as it is not required for this analysis. Disabling SIF simplifies the model and reduces computational complexity and time, focusing the simulation on reflectance spectra and trait parameters. The model state is then initialized using the initial configuration, establishing the baseline state of the model. To ensure consistency during the iterative process of inverse modeling, a deep copy of the initial model state is created as a backup. This allows the model to be reset to its original state as needed during parameter adjustments.

The Leaf Area index (LAI) is an important variable for understanding canopy structure and function, as it determines the optical depth of the vegetation canopy. Due to its dominant influence on surface reflectance, LAI is not calculated through inverse modeling, instead it is derived from the Simple Ratio (SR) of NIR (851-879 nm) to red (636-673 nm) reflectance based on Blinn et al. (2019) for each acquisition date, per pixel (Figure 2):

$$SR = \frac{NIR}{red} \tag{3}$$

with the LAI calculated as:

$$LAI = 0.332915 \times SR - 0.00212 \tag{4}$$

LAI is limited to a practical range of 0.1 to 20 $m^2.m^{-2}$. To establish a common canopy-structure baseline and isolate the effect of leaf traits, we prescribed LAI from SR rather than co-optimizing it with traits. In exploratory tests (Appendix S1), joint LAI–trait inversion was ill-conditioned, with fits collapsing toward LAI-dominated solutions and calculating LAI from SR avoids this instability. Because the coefficients in Eq. (4) are context dependent and canopy properties (e.g., leaf shape, pubescence) can influence SR, we interpret the retrieved quantity as an optical "effective LAI" for radiative-transfer purposes. SR-derived LAI carries a known non-zero offset for bare and senesced surfaces (Baret & Guyot, 1991), therefore retrieved values should be interpreted as total canopy material rather than strictly green leaf area. Any systematic bias introduced by the SR to LAI mapping is common to both the PFT-based and trait-based runs and therefore does not confound their relative comparison. Moreover, photosynthetic capacity is handled via Vcmax25, which scales with leaf chlorophyll content and declines independently of structural LAI as leaves senesce. The clumping index ($\Omega(\theta)$) is another important structural variable that accounts for the horizontal heterogeneity of a pixel within the radiative transfer scheme in CliMA land (Braghiere et al., 2019, 2020, 2021). The clumping index was derived from direct transmittance (P_{gap}) extracted from LiDAR data. Miller (1967) proposed a theorem for the inverse estimation of effective LAI ($LAI_e = LAI \cdot \Omega(\theta)$) that does not require a prior knowledge of the leaf angular distribution function, G(θ) (Ross, 1981). The advantage of using LiDAR P_{gap} values to derive the clumping index is the independence of the viewing zenith angle ($P_{gap}(0°)$), because the sensor points to nadir and uncertainties related to the viewing zenith angle are minimized.

To generate the target reflectance spectrum, the model updates plant trait parameters based on 3 input datasets: AVIRIS-NG hyperspectral surface reflectance (400-2500 nm at 10 nm intervals), LAI, and clumping index (Figure 2). These traits include CHL, LMA, and LWC. For CHL, the parameter range is set between 0.01 and 80 $\mu g.cm^{-2}$, with an initial guess of 10 $\mu g.cm^{-2}$, a step size of 5 $\mu g.cm^{-2}$, and a tolerance of 0.1 $\mu g.cm^{-2}$. For LMA, the range is between 0.1 and 500 $g.m^{-2}$, with an initial guess of 120 $g.m^{-2}$, a step size of 100 $g.m^{-2}$, and a tolerance of 10 $g.m^{-2}$. For LWC, the range is set between 0.1 and 20 $mol.m^{-2}$, with an initial guess of 5 $mol.m^{-2}$, a step size of 1 $mol.m^{-2}$, and a tolerance of 0.1 $mol.m^{-2}$. To facilitate comparison, note that 1 g cm^{-2} equals 1 cm of liquid water; conversions from mol m^{-2} use 18 g mol^{-1} and 10,000 cm^2 m^{-2}. Parameter ranges were selected based on published trait values for Mediterranean vegetation and field observations at Dangermond Preserve, spanning the expected range from herbaceous annuals to evergreen broadleaf trees.

Sensitivity analyses confirmed that < 2% of pixels would exceed these bounds with expanded ranges, indicating the selected parameter space adequately captures trait variability without artificial constraints. These parameters are iteratively updated, and the model recalculates the leaf and canopy level spectra. The spectral response is then matched to reference wavelengths, excluding known water vapor absorption bands (e.g., between 1790-1920 nm and 1345-1415 nm), to generate the target spectrum. Because AVIRIS-NG surface reflectance at ~5 m integrates green leaves with non-photosynthetic vegetation (NPV), soil, litter, and shadow, our LWC represents an apparent canopy-scale LWC and can be biased low in partially vegetated or high-contrast pixels. We document consistency with a Partial Least Squares Regression (PLSR) product (Appendix S1: Figures S2–S3), with field measurements (Appendix S1: Figure S4), and with canopy-scale co-located validation.

The optimization methods used include Reduce Step Method and Solution Tolerance (Kelley, 1999; Nocedal & Wright, 1999), which iteratively adjust the parameters and systematically explore the parameter space to find combinations that minimize the Root Mean Square Error (RMSE) between observed and modeled spectra. The Reduce Step Method is an optimization algorithm that systematically reduces the step size for parameter adjustments as the optimization progresses. This method starts with relatively large steps to quickly explore the parameter space and identify promising regions. As the optimization nears a solution, the step size is gradually reduced to fine-tune the parameter values, ensuring that the algorithm converges to a precise minimum. This method is particularly effective in avoiding local minima and ensuring that the global minimum is found. Solution Tolerance, on the other hand, defines the convergence criteria for the optimization process. It specifies the acceptable level of precision for the parameter values and the maximum number of iterations the solver will perform. For this study, the solution tolerance is set to ensure that the RMSE stabilizes within a specified range across iterations, and the maximum number of iterations is limited to 50. This prevents the solver from running indefinitely and ensures that the optimization process is both efficient and accurate.

2.6 PFT vs. trait-based approaches

In LSMs, spatial heterogeneity is depicted through a structured subgrid approach, where each grid cell may contain multiple PFTs. The number of PFTs varies across models, from as few as 5 in the case of original JULES (Best et al., 2011; Clark et al., 2011) to as many as 22 in Norwegian Earth System Model (NorESM) (Seland et al., 2020), or up to 80 if considering crop types (Lawrence et al., 2019). These PFTs are defined by specific, time invariant parameters that influence a variety of ecological and physiological processes. These include leaf optical properties that affect the leaf's interaction with solar radiation, root distribution parameters that influence water absorption from the soil, aerodynamic characteristics that impact the land-atmosphere transfer of heat, moisture, and momentum, as well as photosynthetic and respiratory parameters that affect stomatal resistance, photosynthesis, and transpiration rates.

While the setup of PFTs within a grid cell generally relies on static configurations, such as using current land cover maps for determining their spatial composition, models capable of dynamic land cover adjustments permit these elements to adapt over time. Importantly, LAI is recognized as a temporally variable parameter within both PFT and trait-based frameworks, necessitating periodic updates to reflect changes. In the domain of CliMA Land, the integration of CHL, LMA and LWC is crucial for the computation of hyperspectral leaf reflectance and transmittance.

In this study, we used two configurations of the CliMA land model: the PFT-based configuration used spectral and photosynthetic leaf properties derived from averaged trait values across time during the SHIFT campaign. Conversely, the trait-based configuration adopts a dynamic approach, utilizing time-variable leaf optical properties (i.e., hyperspectral reflectance and transmittance) and photosynthetic parameters (i.e., V_{cmax25} and J_{max25}) based on time and space varying traits, unlike the fixed parameter setting in the PFT approach, thereby enhancing the model's capability to more accurately capture the temporal and spatial dynamics observed in natural ecosystems (Table 1).

2.7 TROPOMI Solar-Induced Fluorescence

TROPOMI SIF_{740} was gridded to 0.05 degree (~5 km) using a Level-2 binning routine over the Dangermond area of interest (34.4405°–34.5847° N, 120.508°–120.333° W) for 2022-02-01 to 2022-06-30. CliMA Land outputs of 30 m were aggregated to the same 0.05° grid by computing the mean of all valid 30 m pixels whose center coordinates fell within each 0.05° target grid cell, thereby including all contributing model pixels within the footprint. For evaluation, analyses were restricted to grid cells whose centers lie within the Preserve polygon (land only), thereby excluding open-ocean cells and reducing coastal mixing. Following spatial averaging, SIF values were temporally averaged over a period of 9 days with a range of four days before and after the target date (SHIFT flights). This temporal averaging approach involved selecting SIF data within a specific time range centered around the target date has been previously demonstrated to enhance the signal's reliability (Turner et al., 2020). The duration of one week with a four-day buffer provides a broader temporal context for analysis, capturing potential fluctuations in SIF values over time. In comparison, CliMA Land SIF was modeled at nadir direction (viewing zenith angle is 0°), and processed identically. To enable a physically consistent comparison, both model and satellite SIF were expressed as relative SIF, defined as the ratio of SIF_{740} to the background continuum radiance at 740 nm, multiplied by 100 to yield units of percent. For TROPOMI, this quantity is provided directly in the Level-2 product, defined as the fluorescence signal relative to the spectral continuum estimated from the 743–758 nm fitting window (Köhler et al., 2018; Joiner et al., 2020). For CliMA Land, relative SIF was computed as SIF_{740} divided by the total upwelling radiance at 740 nm. TROPOMI observations include both positive and negative retrievals due to instrument noise; following Joiner et al. (2020), all valid retrievals were retained in spatial averages since excluding negative values introduces a positive bias in the mean.

3. Results

3.1 Seasonal Dynamics of Plant Traits

Figure 3a illustrates the temporal dynamics and seasonal growth patterns of LAI for each PFT within the Dangermond Preserve, demonstrating how these patterns reflect the plants' adaptations to the Mediterranean climate's seasonality. PFT 1, the Coastal Scrub, displays a decline in LAI from a peak of 1.81 $m^2.m^{-2}$ to a minimum of 0.98 $m^2.m^{-2}$, with a standard deviation (STD) of 0.306, from early spring to early summer. This pattern is aligned with the survival strategies of these plants, which involve conserving water during the less moist conditions of late spring and early summer. For PFT 2, the Annual Grassland, a marked decline in LAI is observed as the seasons change, starting from a high of 1.93 $m^2.m^{-2}$ and reaching the same minimum as PFT 1 of 0.88 $m^2.m^{-2}$, with a higher STD of 0.375. This trend is indicative of the rapid vegetative growth during the wet season and the subsequent lifecycle completion during the dry season, which contributes to the ecosystem through carbon sequestration and nutrient recycling upon decay.

PFT 3, the Coastal Oak Woodland, presents the most significant fluctuation in LAI values, ranging from a maximum of 2.27 $m^2.m^{-2}$ to a minimum of 1.44 $m^2.m^{-2}$, with a STD of 0.282. The seasonal amplitude (max–min) is comparable across PFTs (≈ 0.83–1.06 $m^2.m^{-2}$), with grassland showing the largest amplitude and oak the smallest, while the temporal STD indicates greater intra-season variability in grasslands.

The initial high LAI values reflect the dense canopies formed during the favorable conditions of the wet season, with a mid-season drop and a recovery phase potentially indicating a response to rain events along the campaign (see Appendix S1). The resilience in LAI values towards the end of the period may be attributed to the oaks' deep root systems, which enable them to better withstand drought conditions.

Figure 3b and Figure 3c present the temporal dynamics of CHL and LMA across different PFTs in the Dangermond Preserve, respectively. These figures show how each PFT adapts to the cyclical wet and dry periods characteristic of this climate zone. PFT 1 experiences a significant decline in CHL (~68%), starting from a high of 46.7 $\mu g.cm^{-2}$ and ending at a minimum of 14.7 $\mu g.cm^{-2}$, with the variability suggesting the influence of the drying conditions of late spring and early summer on photosynthetic activity. PFT 2's CHL also decreases over time from an initial 42.0 $\mu g.cm^{-2}$ to 11.0 $\mu g.cm^{-2}$, reflecting similar stresses due to season changes. PFT 3's CHL decreases from 61.1 $\mu g.cm^{-2}$ to 25.2 $\mu g.cm^{-2}$ (~59%), indicating a better adaptation mechanism that allows for more consistent photosynthetic performance throughout the season. These values are in alignment with *in-situ* data (N= 183; bias = 8.78 $\mu g.cm^{-2}$; Appendix S1: Figure S4) collected during the campaign (Chadwick et al., 2025).

The LMA values offer insights into the structural adaptations of the plants' leaves. PFT 1 shows a moderate fluctuation in LMA, peaking at 79 $g.m^{-2}$ and dropping to 25 $g.m^{-2}$, indicating a response to the environmental conditions while maintaining leaf structure to a certain extent. PFT 2 demonstrates a change in LMA, with values ranging from 68 g.m^{-2} to 20 g.m^{-2}, which suggest structural changes in leaf density or thickness as the dry season intensifies. PFT 3, with an LMA range from 151 g.m^{-2} to 52 g.m^{-2}, shows higher overall values, implying a higher level of resilience to the seasonal dynamics.

The LWC values illustrate the temporal dynamics of leaf water among different PFTs (Figure 3d). PFT 1 exhibits a gradual decline in LWC, starting around 0.0057 g.cm^{-2} and stabilizing at approximately 0.0012 g.cm^{-2}, indicating a consistent reduction in leaf water content. PFT 2 shows a similar pattern, with LWC values decreasing from about 0.0053 g.cm^{-2} to just above 0.00092 g.cm^{-2}. PFT 3 experiences a significant decline in LWC, beginning at around 0.0100 g.cm^{-2} and dropping to below 0.0026 g.cm^{-2}.

3.2 Plant trait variability within PFTs

As plant traits vary with the seasons, they also vary within PFTs. This variability is a key factor in understanding the adaptive strategies of plants to environmental fluctuations and stressors across different species and in space. Figures 4–6 show seasonal histograms, early (late Feb–Mar; Figure 4), mid (late Mar–Apr; Figure 5), and late (May; Figure 6), for each trait and PFT, which reduces time-averaging effects and makes phenological differences clearer.

For CHL in $\mu g\ cm^{-2}$: early-season means (±STD) were 42.75 ± 21.75 (PFT1), 38.21 ± 20.23 (PFT2), and 57.90 ± 19.22 (PFT3). By late spring they declined to 17.64 ± 13.83, 13.66 ± 11.44, and 29.50 ± 16.81, corresponding to decreases of 58.7%, 64.2%, and 49.0% from early to late. The early-to-mid reductions were 26.8%, 29.5%, and 20.5% for PFT1–PFT3, followed by mid-to-late reductions of 43.7%, 49.3%, and 35.9%. Variability also tightened from early to late

(STD change: −36.4%, −43.5%, −12.5% for PFT1–PFT3), consistent with the narrowing of states as senescence progresses.

For LMA in g m^{-2}: means decreased from 68.67 ± 76.29 (PFT1), 58.94 ± 68.54 (PFT2), and 130.55 ± 98.73 (PFT3) in early spring to 28.65 ± 34.71, 23.73 ± 31.60, and 58.11 ± 42.45 in late spring, corresponding to declines of 58.3%, 59.7%, and 55.5%, respectively (early to late). Dispersion contracted substantially over the season, with STD reductions of 54.5% (PFT1), 53.9% (PFT2), and 57.0% (PFT3).

For LWC in g cm^{-2} (values shown $\times 10^{-3}$): means fell sharply from early to late: 5.04 ± 5.70 to 1.42 ± 2.23 for PFT1 (−71.9%), 4.66 ± 5.40 to 1.14 ± 2.05 for PFT2 (−75.6%), and 8.99 ± 7.24 to 3.01 ± 2.54 for PFT3 (−66.5%). The early-to-mid and mid-to-late steps were −37.3% and −55.1% (PFT1), −36.3% and −61.7% (PFT2), and −33.8% and −49.4% (PFT3). Variability collapsed across PFTs (STD change: −61.0%, −62.0%, −64.9%), reflecting progressive canopy desiccation and convergence to low-water states by late spring. These magnitudes are independently corroborated by the AVIRIS-NG PLSR product (Figs. S2–S3) and fall within the field range (N = 222; bias = −0.041 g cm^{-2}, RMSE = 0.113 g cm^{-2}; Appendix S1: Figure S4). The apparent low bias in LWC relative to raw field measurements is consistent with canopy-scale sub-pixel mixing and retrieval-scale mismatch, as shown by canopy-integrated validation (see Appendix S1).

CHL and LWC decline markedly from early to late spring, with the largest reductions in annual grassland and scrub, while oaks maintain higher central tendencies throughout. LMA also decreases substantially but preserves the expected structural ranking among PFTs. These results indicate that early-season overlap arises from shared green, well-watered conditions, whereas late-season divergence reflects differential senescence and water loss across PFTs.

In addition to the baseline inversion used in the main analysis, we also examined how the recovered trait distributions depend on which traits and structural variables are optimized in CliMA Land. Specifically, we repeated the retrievals (i) jointly fitting LAI, CHL and LMA, and (ii) fitting CHL, LMA and LWC under a simplified, horizontally homogeneous canopy ($\Omega = 1$), and compared these to the vegetation-index–based trait products (Appendix S1: Figures S5–S13). Across these configurations, absolute trait magnitudes shift somewhat, but the cross-PFT ordering (oaks > scrub/grass for LMA and LWC, higher CHL in oaks early in the season) and the large within-PFT spreads are preserved. Spatial patterns also remain highly correlated. These sensitivity tests show that our main conclusions about strong temporal variation and substantial heterogeneity of traits within each PFT are robust to the specific choice of optimized trait set and canopy-structure assumptions.

3.3 Spatial variability between plant traits and PFT approaches

Figures 7 and 8 show the spatial distribution and variability CHL, LMA, and LWC within Dangermond Preserve, highlighting the role of biophysical and topographical variation in influencing these plant traits. The analysis of time-averaged hyperspectral data from the SHIFT campaign captures a comprehensive picture of these biophysical parameters.

The spatially explicit trait maps in the left column of Figure 7 display a high degree of variability within the preserve, with higher concentrations CHL, LMA, and LWC typically found in the higher terrain, and often associated with the coastal oaks. These areas exhibit elevated trait values, suggesting a strong influence of topography (see Appendix S1: Figures S14–S15). The right column of Figure 7 offers a visualization of average PFT values, yet this consolidation masks the nuanced variations observed in the detailed trait maps.

Comparative analysis of the average trait values against the PFT averages provides insights into the distribution patterns across the preserve. While PFT categorization provides useful generalizations, the trait-specific maps yield finer granularity that is important for understanding ecological processes.

Figure 8, quantifying the average differences (trait – PFT), offers an illustration of how topography can modulate vegetative traits (see topography in Figure 1 and Appendix S1). These differences highlight the limitations of using PFTs as a sole representation of ecosystem functionality. By factoring in the elevation-induced disparities, the trait-based approach offers a dynamic perspective, indicating that topographical and species-specific features play a significant role in water, light, and nutrient availability, thus influencing CHL, LMA, and LWC levels.

3.4 Impact of Trait Variability on Photosynthesis and SIF Modeling

Figures 9 and 10 offer an examination of the disparities in GPP and SIF_{740} predictions made by the trait-based and PFT-based approaches within the CliMA Land model across the Dangermond Preserve. These differences are important for understanding how each modeling approach reflects the complex interplay between vegetation, topography, and soil water availability during the SHIFT campaign.

Figure 9a and Figure 9b represent the difference in predicted values, with positive (red) indicating where the trait-based model predicts higher GPP and SIF values, respectively, compared to the PFT-based model. These differences indicate the magnitude of deviation between the two modeling approaches. Similarly to Figure 8, the red areas of the maps are concentrated along slopes and elevations, suggesting that the trait-based model is more sensitive to fine-scale environmental variations that influence plant physiology.

The spatial variability observed underscores that plant traits are deeply intertwined with the terrain. The positive mean and significant STD values (0.31 ± 2.52 μmol CO_2 m^{-2} s^{-1} for GPP and 0.006 ± 0.028 mW m^{-2} sr^{-1} nm^{-1} for SIF_{740}) provided for the differences in GPP and SIF means that while the trait-based approach results in overall higher values of productivity and SIF, there is also considerable heterogeneity within the preserve related mostly with species composition and topography. This variability is likely driven by the complex interactions between vegetation characteristics and the localized environmental factors that the trait-based approach more finely resolves.

The time series in Figure 10a and Figure 10b track the temporal progression of GPP and relative SIF_{740} from February to June 2022. The trait-based predictions (blue) do not uniformly overestimate the PFT predictions (orange); instead, they exhibit higher predicted values at the beginning of the season and lower at the end. This trend indicates that the trait-based approach, with its underlying assumptions and parametrizations, yields higher estimates for these two indicators of plant activity when water is not a limiting factor. The oscillations in the trait-based approach likely mirror the natural responses of plant physiology to seasonal environmental changes, which may include shifts in temperature, light availability, and water stress.

Figure 10b serves as a quantitative bridge, connecting the model's predictive capacity with real-world satellite observations. The evaluation of CliMA Land's output against TROPOMI relative SIF_{740} data (%) presents a direct comparison, highlighting the nuances in the relationship between the models' predictions and the observed SIF values and the predictive performances of the PFT and trait approaches. The regression analysis for both approaches suggests that each model has similar explanatory power for the variance in observed SIF data, while the trait-based approach performs marginally better. The PFT-based and trait-based approaches showed comparable explanatory power against TROPOMI relative SIF_{740} with $R^2 = 0.68$ and $R^2 = 0.69$, respectively. R^2 is computed against the per-date ±4-day TROPOMI means (13 SHIFT dates). A systematic offset of approximately 20 – 25% remained between modeled and TROPOMI relative SIF_{740}, particularly early in the season. Spatial agreements are shown in Appendix S1: Figure S16. Over the 94-day, 9,903-ha domain, Trait accumulated ~5% more CO_2 uptake than PFT (see Appendix S1: Figure S17 for magnitudes and STD).

4. Discussion

The study conducted at Dangermond Preserve using NASA's SHIFT campaign data provides a case for the integration of trait-based approaches in LSMs. The evidence gathered and analyzed shows an improvement in the representation of ecosystem dynamics when high-resolution, time-variant plant trait data is utilized over the conventional PFT based approach.

The findings suggest that the trait-based approach produced a marginal improvement in TROPOMI relative SIF_{740} agreement (Figure 10b), but it changed the seasonal trajectory and spatial distribution of modeled GPP and SIF, which impacts calculations of carbon uptake. A systematic offset (~20–25%) remains between model and TROPOMI relative SIF_{740}. This likely reflects a combination of factors including that the model uses total scene radiance while TROPOMI's spectral retrieval is less sensitive to spectrally smooth soil contributions (Köhler et al., 2018), as well as unaccounted bidirectional effects arising from TROPOMI's wide-swath viewing geometry (Joiner et al., 2020). Since terrestrial ecosystems play a crucial role in the global carbon cycle, more precise modeling approaches can better capture vegetation heterogeneity and their responses to the environment can potentially provide better predictions.

At the same time, a lack of spatially explicit trait data may increase the parametric uncertainty of these models making it harder to perform the same type of study globally. Therefore, more surface hyperspectral reflectance data at high spatial and temporal resolutions are needed because the variability in traits, and their influence on photosynthesis, are critical for understanding ecosystem responses to climate change (Fernández-Martínez et al., 2020; Green et al., 2022).

It is important to highlight that the PFT designations used in our study are significantly more specific and detailed than those typically employed in regional-to-global studies. Often, such studies abstract PFTs into very generic categories, such as "C3 grass" or "temperate broadleaf tree". By using more precise PFT classifications, our trait-based approach likely provides more accurate and insightful results. Consequently, any benefits demonstrated here from utilizing trait-based methods are probably underestimated when compared to the advantages over the more generic PFTs commonly used in many modeling studies. This underscores the value of detailed trait-based analysis in improving ecological modeling and predictions.

Likewise, it is important to distinguish between two types of models concerning LAI. Some models treat LAI as an externally prescribed "time-varying parameter" or "driver," while others consider LAI as an internally evolving state variable that estimates changes over time, such as in most dynamic global vegetation models (DGVMs). This distinction is fundamentally separate from the trait-based versus PFT-based approach. Whether LAI is prescribed or simulated is independent of whether models use discrete trait values by PFT or continuously varying traits. In this study, LAI was prescribed as a time-varying driver derived from AVIRIS-NG reflectance for each 30-m grid cell, SR was computed and converted to LAI, yielding a temporally varying LAI field over the campaign. Recognizing this distinction helps clarify the different modeling frameworks and their implications for ecological predictions and analysis.

A modeling choice in this study was to prescribe LAI from remote sensing rather than co-optimize LAI with traits. Our objective was not to benchmark LAI retrievals per se but to test whether trait variation improves flux estimates relative to a fixed-trait PFT baseline under a shared LAI forcing. While SR captures the red-edge contrast linked to canopy structure, cross-PFT variability (e.g., leaf shape, pubescent leaves, among others) can bias absolute LAI. We therefore interpret these values as optical “effective LAI” for radiative-transfer purposes and emphasize relative differences between approaches. Future work should incorporate PFT-specific SR to LAI calibrations and/or direct LAI constraints (e.g., co-located field plots) to refine absolute magnitudes.

The high correlation among retrieved traits reflects mechanistically coupled canopy processes rather than an optimization artifact. As vegetation responds to declining water availability, LAI decreases through leaf shedding and senescence, CHL degrades, and LWC declines, processes that are physiologically linked through shared drought responses. This covariation is both expected and ecologically meaningful. However, it does present numerical challenges for simultaneous multi-trait optimization, as the dominant influence of LAI on canopy reflectance can impact spectral variation that should be attributed to other traits. Our approach of prescribing LAI while retrieving CHL, LMA, and LWC avoids this ill-conditioning while maintaining focus on the spatial and temporal trait variability that distinguishes the trait-based from PFT-based approach.

This study also highlights the importance of biodiversity representations in climate models because traditional PFT-based models often simplify the vast diversity of plant responses to their environments, which can lead to significant gaps in the accuracy of these models. By incorporating diverse plant traits that vary seasonally and spatially among different species within the same functional type, trait-based approaches can offer a more realistic and nuanced representation of ecosystem functions (Hakkenberg et al., 2021; Harrison et al., 2020).

The seasonal dynamics of plant traits evaluations emphasize the importance of considering the seasonal growth patterns and physiological changes of plants in different PFTs when modeling ecosystem functions. The retrieved trait dynamics are supported by field measurements collected during the SHIFT campaign (Appendix S1: Figure S4), which show that our inversions capture realistic trait magnitudes and overall seasonal patterns. The decline in CHL during the transition to the dry season indicates a potential reduction in growth rates and photosynthetic capacity, which is vital for predicting carbon and water fluxes in these ecosystems. Such detailed temporal analysis is important for enhancing the accuracy of LSMs, particularly for projections in arid regions with Mediterranean climates where water availability and temperature shifts play a critical role in ecosystem dynamics. However, our approach relies on empirical trait–photosynthesis relationships that may introduce uncertainty. The Vcmax–CHL relationships from Croft et al. (2017), derived from North American forests, may not be optimally calibrated for Mediterranean vegetation types, though this limitation applies equally to both modeling approaches and does not affect our primary finding that trait-based parameterization better captures spatial and temporal heterogeneity than PFT-based approaches. A limitation of our analysis is that it is restricted to the 13-week SHIFT campaign window from late winter to late spring, and therefore does not encompass the full annual cycle of GPP and SIF at Dangermond Preserve. This constraint, however, reflects the design of the intensive airborne–ground campaign, which was optimized to sample the rising limb and peak of seasonal activity with repeated AVIRIS-NG flights and coordinated ground observations, rather than an oversight of the study. Extending trait-based modeling to full annual cycles will require complementing such intensive campaigns, which we identify as an important avenue for future work.

The overlapping areas of Figures 4–6 suggest that while there may be average differences in traits among PFTs, individual plants within each PFT show an overlap in their traits, especially in early spring when canopies are green and well-watered. As the season progresses, the season-resolved panels reveal clearer separation (e.g., CHL and LWC shifting lower in grasses and scrub while oaks retain higher central tendencies). This indicates that external factors such as microclimate conditions, soil nutrition, water availability, plant age or health, and even uncertainties in the PFT classification map could be influencing traits within these groups. Correlations between topographic variables (elevation, slope, aspect) and optimized traits across the study area are shown in Appendix S1: Figures S14 and S15. Elevation showed significant positive correlations with CHL ($r = 0.39$, $p < 0.001$), LMA ($r = 0.17$, $p < 0.001$), and LWC ($r = 0.11$, $p < 0.001$). Slope exhibited moderate correlations with all three traits (CHL: $r = 0.34$, LMA: $r = 0.23$, LWC: $r = 0.18$; all $p < 0.001$), while aspect showed weak but significant relationships ($r = 0.01$–0.03, $p < 0.05$). These results suggest that topography-related environmental gradients, including temperature (elevation ~231 ± 119 m), water drainage (slope ~26 ± 17°), and solar exposure (aspect), contribute to the observed within-PFT trait variability. Such diversity within PFTs challenges the assumption that a single set of trait values can accurately represent the functional group in its entirety. Instead, it highlights the need for LSMs to incorporate a spectrum of trait expressions to refine predictions of photosynthesis, growth, and carbon cycling across different vegetation types and environmental conditions.

The CHL histograms are unimodal with mild right-skew, consistent with a trait that adjusts plastically within season (Figs. 4a, 5a, 6a). LMA exhibits a broad, heavy-tailed (approximately log-normal) distribution (note the log y-axis; Figs. 4b, 5b, 6b), reflecting structural differences among species and leaf habits within each PFT. LWC is strongly right-skewed and a sparse high tail (Figs. 4c, 5c, 6c). Together, these shapes indicate that CHL varies rapidly with phenology, LMA is more phenotypically conserved, and LWC integrates both structure and water status, making it sensitive to senescence and cover.

The spatially explicit trait maps (Figure 7) show that CHL, LMA, and LWC are strongly organized along terrain gradients rather than uniformly distributed within PFTs. CHL exhibits the clearest signal, increasing systematically with elevation and slope, forming areas that coincide with oak woodlands and on dissected hillslopes. This pattern likely reflects both community composition and topography-driven microclimatic gradients in light and temperature that favor higher photosynthetic capacity. Overall, these patterns indicate that topography exerts a first-order control on trait expression at landscape scale.

In interpreting correlations between CHL and LMA, it is important to consider the potential artifacts introduced by area versus mass normalization. Positive correlations observed between these traits might not purely reflect biological relationships but could be influenced by the mathematical effects of normalization. Osnas et al. (2013) discuss these issues, highlighting how normalizing by area or mass can create spurious correlations. Understanding these nuances can be important for advancing the knowledge of trait variability and PFTs in ecological studies.

The spatial variability observed in GPP and SIF underscores the deep interconnection between plant traits and terrain. The positive mean and significant standard deviation values indicate that the trait-based approach results in higher overall productivity and SIF but also considerable heterogeneity within the preserve, primarily due to species composition and topography. Currently, all plants in this site experience identical ERA5 reanalysis inputs. However, in practice, local-scale differences in temperature, radiation (e.g., north-facing vs. south-facing slopes), and water availability driven by topography, exist. Given that simulated GPP and SIF responses are strongly structured by terrain, it is likely that a more pronounced modification of GPP would be observed if these local-scale differences were accounted for in the model. Resolving fine-scale topography and incorporating localized environmental variability would provide a more accurate representation of plant responses and further enhance the predictive power of the trait-based approach.

While the trait-based approach presents clear advantages in small scale studies, it also poses significant challenges, particularly in terms of data collection, computational demands, and parameter uncertainty. These challenges are amplified when integrating remote sensing-based estimates of plant traits. For instance, the correct ecological interpretation of a 5m trait map can be conceptually challenging. Often, remote sensing-based plant traits are considered a community-weighted mean of sunlit leaves, but this raises issues. Firstly, it is not directly clear how representative sunlit leaves are of the entire leaf population. Secondly, it is difficult to determine whether the community-weighted mean actually captures trait variation or species variation. Model-based inference and prediction approaches (Clark, 2016) highlight these intricacies by using species composition data to derive model-based estimates or predictions of trait covariances. These approaches account for both continuous and categorical variables, as well as ordinal traits like shade, drought, and flood tolerance, demonstrating the necessity of robust modeling frameworks to understand how species distribution and environmental factors influence trait diversity.

This current approach also requires frequent, high-quality hyperspectral observations with adequate temporal coverage, which are increasingly available through global platforms such as EMIT (Green et al., 2020; Thompson et al., 2019, 2024), EnMAP (Guanter et al., 2015), PRISMA (Cogliati et al., 2021), GEDI (Dubayah et al., 2020), and PACE (Werdell et al., 2019). Field campaigns and observatory networks like NEON (Kampe et al., 2010) and BioSCape (Cardoso et al., 2025) provide critical calibration and validation data across biomes. Remotely sensed traits represent community-weighted means (Garnier et al., 2004; Violle et al., 2007) that integrate multiple processes, including individual-scale plasticity, evolutionary adaptation, and community compositional shifts (Clark et al., 2011; Kattge et al., 2020). While this integration makes trait-based approaches valuable for current and near-term predictions where observational constraints exist, longer-term projections for novel future conditions require models that mechanistically represent the processes governing trait variation (Serbin et al., 2015; Singh et al., 2015), including optimality theory (Dong et al., 2023; Wang et al., 2023). Therefore, trait-based and PFT-based approaches are complementary research directions rather than competing frameworks.

High-resolution hyperspectral imaging and the processing of large datasets require substantial resources and sophisticated modeling techniques. Future research should focus on optimizing these methods and exploring their applicability across different biomes and scales, including the use of new space missions (Cawse-Nicholson et al., 2021) and machine learning (Cai et al., 2023), GPU native models (Deck et al., 2026), as well as systematically investigate environmental controls on trait variation, including topography, soil moisture, and microclimate, to develop process-based trait parameterizations that could replace static PFT assignments in LSMs. Furthermore, integrating this approach into LSMs requires collaborative efforts among scientists to standardize trait measurements and model formulations, ensuring that they can effectively contribute to comprehensive ecosystem and climate projections.

5. Conclusion

This study focused at the Dangermond Preserve, utilizing high-resolution hyperspectral and LiDAR data from NASA's SHIFT campaign, demonstrates advancements in the representation of ecosystem function through the integration of plant traits into the CliMA Land model. Our findings underscore the limitations inherent in traditional PFT-based approaches within LSMs and highlight the enhanced ecological fidelity offered by a trait-based modeling approach.

Despite the benefits, the adoption of trait-based models poses challenges, primarily related to the increased demands for detailed, high-quality spectral data (now increasingly available globally) and the computational resources required for processing such information. Future research should therefore focus on optimizing data collection, enhancing model capabilities and efficiency, as well as extending the applicability of trait-based approaches across ecosystems globally. This includes harnessing advancements from forthcoming space missions and leveraging innovative technologies such as machine learning to further refine the accuracy and scalability of these models, as well as mechanistic understanding of trait variation processes for application to future projections beyond observed conditions.

The incorporation of time-variant, high-resolution plant traits into leaf optical and physiological parameterizations improved the ecological fidelity of CliMA Land simulations across the heterogeneous landscapes of the Dangermond Preserve. Although the trait-based configuration produced a marginal improvement in coarse-scale TROPOMI relative SIF_{740} agreement, it better reflected the spatial heterogeneity, and phenological strategies of vegetation.

Furthermore, comparative analysis of the trait-based and PFT-based methods reveals that the former yields improved representations of phenology and spatial heterogeneity. Specifically, it captures higher carbon uptake and sharper relative SIF_{740} changes during key seasonal transitions, suggesting that traditional PFT methods may underrepresent critical aspects of carbon cycling.

In conclusion, integrating high-resolution spectral observations, repeated temporal sampling, and trait-based modeling approaches into LSMs offers a promising avenue for enhancing our understanding of ecosystem responses to climate change. It is a step forward in bridging the gap between ecological complexity and climate modeling.

Acknowledgments

Part of this research was carried out at the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration (80NM0018D0004). We acknowledge high-performance computing support from Caltech. This work was supported in part by the Resnick Sustainability Institute at Caltech. Part of this research was funded by Schmidt Sciences, LLC.

Author Contributions

R.K.B. conceived the study, led the analysis, and wrote the manuscript. Y.W. and K.L. helped with the CliMA Land simulations and TROPOMI data. K.D.C., P.G.B., A.N.S., D.S. led data collection and preprocessing for the SHIFT campaign and derived trait products. F.D.S. and A.F. contributed to model–data integration and interpretation. T.Z., N.Q., and P.A.T. provided field and PLSR data. C.F. supervised the project and contributed to study design and manuscript editing. All authors discussed the results and contributed to the final manuscript.

Conflict of Interest Statement

The authors declare no conflict of interest.

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Table 1: Comparative Analysis of Vegetation Modeling Approaches in CliMA Land: PFT vs. Trait-Based

Feature	**PFT Approach**	**Trait-Based Approach**
Number of PFTs	3 (scrub, grass, oak)	Utilizes high-resolution plant

		trait maps (30 m)
Leaf Optical Properties (hyperspectral reflectance & transmittance)	Static, parameters defined for each PFT based on average CHL, LMA, and LWC	Dynamic, influenced by time varying CHL, LMA, and LWC
Photosynthetic Parameters (Vcmax25 & Jmax25)	Static, parameters defined for each PFT based on average CHL, LMA, and LWC	Dynamic, influenced by time varying CHL, LMA, and LWC
Structural Parameters (LAI)	Dynamic, time-varying per pixel, derived from AVIRIS-NG	Dynamic, time-varying per pixel, derived from AVIRIS-NG

Figure captions

Figure 1. The Jack and Laura Dangermond Preserve at Point Conception with three dominant Plant Functional Types (PFTs) classified according to the California Vegetation – WHRTYPE categories: i. Coastal Scrub (green), ii. Annual Grasslands (blue), and iii. Coastal Oak Woodland (red). The inset photos provide a visual reference to the distinct landscapes represented by each PFT. The shading illustrates the complex topography of the region, based on hillshades of the Dangermond Preserve 2 m digital terrain model inside the preserve, and the USGS CA SoCAL Wildfires 2018 D18 digital terrain model at 1 ⁄ 3 arc-second outside of the preserve. Photo credit: R. K. Braghiere (2022).

Figure 2. Schematic illustrating the workflow of the CliMA Land model used for estimating ecosystem traits and fluxes. The process involves the integration of remote sensing data from AVIRIS-NG and LiDAR Clumping Index (1) to inform the model with target hyperspectral surface reflectance. Inverse modeling (2) is employed to derive trait variables leaf chlorophyll content per area (CHL), leaf mass per area (LMA), and leaf water content (LWC). These traits

are then used in forward modeling (3) with ERA5 reanalysis data to estimate (4) GPP (Gross Primary Productivity) and SIF (Solar-Induced Fluorescence) fluxes in forward modeling. The resulting spatial maps provide insights into the mean and standard deviation (STD) of both traits and fluxes across the Dangermond Preserve during NASA's SHIFT campaign.

Figure 3. Seasonal Dynamics of **a.** Leaf Area Index (LAI), **b.** leaf chlorophyll content (CHL), **c.** leaf mass per area (LMA), and **d.** leaf water content (LWC) with interquartile range shading across Plant Functional Types (PFTs) in the Dangermond Preserve during NASA's SHIFT campaign (PFT 1 = Coastal Scrub, PFT 2 = Annual Grassland, and PFT 3 = Coastal Oak Woodland). The constant lines represent the average values across time per PFT, used as the reference values per PFT. Panel **a.** includes ERA5 precipitation and temperature.

Figure 4. Histograms of the Early (late Feb–Mar 2022) AVIRIS-NG acquisition for (a) leaf chlorophyll content (CHL, $\mu g\ cm^{-2}$), (b) leaf mass per area (LMA, $g\ m^{-2}$), and (c) leaf water content (LWC, $\times 10^{-3}\ g\ cm^{-2}$). Colors denote Plant Functional Types (PFTs) (PFT 1 = Coastal Scrub, PFT 2 = Annual Grassland, and PFT 3 = Coastal Oak Woodland). Dashed lines mark PFT means.

Figure 5. Histograms of the Mid (late Mar–Apr 2022) AVIRIS-NG acquisition for (a) leaf chlorophyll content (CHL, $\mu g\ cm^{-2}$), (b) leaf mass per area (LMA, $g\ m^{-2}$), and (c) leaf water content (LWC, $\times 10^{-3}\ g\ cm^{-2}$). Colors denote Plant Functional Types (PFTs) (PFT 1 = Coastal Scrub, PFT 2 = Annual Grassland, and PFT 3 = Coastal Oak Woodland). Dashed lines mark PFT means.

Figure 6. Histograms of the Late (May 2022) AVIRIS-NG acquisition for (a) leaf chlorophyll content (CHL, $\mu g\ cm^{-2}$), (b) leaf mass per area (LMA, $g\ m^{-2}$), and (c) leaf water content (LWC, $\times 10^{-3}\ g\ cm^{-2}$). Colors denote Plant Functional Types (PFTs) (PFT 1 = Coastal Scrub, PFT 2 = Annual Grassland, and PFT 3 = Coastal Oak Woodland). Dashed lines mark PFT means.

Figure 7. Spatial distribution of leaf traits across Dangermond Preserve, derived from time-averaged hyperspectral data during the SHIFT campaign: (a) trait-based and (b) Plant Functional Type (PFT)-based leaf chlorophyll content (CHL); (c) trait-based and (d) PFT-based leaf mass per area (LMA); and (e) trait-based and (f) PFT-based leaf water content (LWC). Trait-based maps (a, c, e) are spatially explicit; PFT-based maps (b, d, f) show values averaged within each PFT (PFT 1 = Coastal Scrub, PFT 2 = Annual Grassland, and PFT 3 = Coastal Oak Woodland).

Figure 8. Average differences between trait-based and Plant Functional Type (PFT)-based approaches (trait – PFT) for (a) leaf chlorophyll content (CHL), (b) leaf mass per area (LMA), and (c) leaf water content (LWC) across the Dangermond Preserve, highlighting the within-landscape variations masked by PFT averaging.

Figure 9. Spatial distribution of differences between trait-based and Plant Functional Type (PFT)-based CliMA Land simulations for (a) gross primary production (GPP, μmol CO_2 m^{-2} s^{-1}) and (b) solar-induced fluorescence at 740 nm (SIF_{740}, mW m^{-2} sr^{-1} nm^{-1}) across the Dangermond Preserve during the SHIFT campaign. Red areas indicate where the trait-based simulations predict higher values than the PFT-based simulation; blue areas indicate the opposite. Insets show the distribution of trait minus PFT differences.

Figure 10. Time series of (a) gross primary production (GPP) and (b) relative solar-induced fluorescence at 740 nm (SIF_{740}) values from February to June 2022, calculated using the trait (blue) and Plant Functional Type (PFT) (orange) approaches in CliMA Land. In panel (b), model relative SIF_{740} (left axis) is compared against TROPOspheric Monitoring Instrument (TROPOMI) observations (right axis; grey triangles show ± 4-day averages, black line shows a Gaussian-smoothed mean ± standard error of the mean (SEM)).